

MAT1002 Lecture 15,

Outline

- Chain rule
 - ↳ Implicit differentiation
- Directional derivatives and gradient vectors

Chain Rule

Motivation: knowing $\frac{dw}{dt}|_{t=t_0}$ and $\frac{dl}{dt}|_{t=t_0}$, what is $\frac{dA}{dt}|_{t=t_0}$ for a rectangle?

Special Case 1 $f(x(t), y(t))$.

Suppose $z = f(x, y)$, $x = x(t)$, $y = y(t)$, all functions involved are differentiable. Then $z = f(x(t), y(t))$ is differentiable, and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

Example

Suppose that $z = x^2y + 3xy^4$, $x = \sin(2t)$ and $y = \cos t$. Find dz/dt when $t = 0$.

Sol: $\frac{dz}{dt}|_{t=0} = 6.$

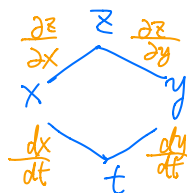
Proof of Special case: (Outline)

- Fix $t = t_0$; introduce Δt . Consider Δz , Δx and Δy caused by Δt .
- By differentiability (and hence continuity), $\Delta x, \Delta y \rightarrow 0$ as $\Delta t \rightarrow 0$.
- Write out Δz as a function of Δx and Δy using

definition of differentiability.

- Take Δt to obtain the desired formula. □

"Branch diagram":



General Case 1 $f(x_1(t), x_2(t), \dots, x_n(t))$.

Suppose $w = f(x_1, \dots, x_n)$, $x_i = x_i(t), \dots, x_n = x_n(t)$, all differentiable. Then w is differentiable w.r. to t , and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x_1} \frac{dx_1}{dt} + \dots + \frac{\partial w}{\partial x_n} \frac{dx_n}{dt}.$$

Here x_i is the i th variable of f .

Q: How if $z = f(x, y)$, but x and y are both 2-variable functions, say $x = x(s, t)$ and $y = y(s, t)$? E.g., z is altitude of mountain, by you want to know $\frac{\partial z}{\partial \theta}$, where $x = r \cos \theta$ and $y = r \sin \theta$?

Special Case 2

Suppose $z = f(x, y)$, $x = x(s, t)$, $y = y(s, t)$, all diff'able.

Then z is diff'able in (s, t) , and

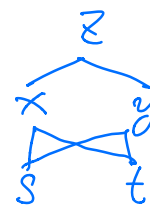
$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}.$$

Example

Suppose that $z = e^x \sin y$, $x = st^2$ and $y = s^2t$. Find $\partial z / \partial s$ and $\partial z / \partial t$.

Sol: $\frac{\partial z}{\partial s} = \sin(s^2t) e^{st^2} t^2 + \cos(s^2t) e^{st^2} \cdot 2st$

$\frac{\partial z}{\partial t}$: exercise.

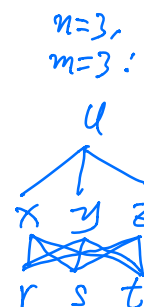


General Case

Theorem (Chain Rule)

Suppose that $u = f(x_1, x_2, \dots, x_n)$ is a differentiable real-valued function with n variables, and suppose that for each i , $x_i = g_i(t_1, t_2, \dots, t_m)$ is a differentiable real-valued function with m variables. Then u is differentiable with respect to $(t_1, t_2, \dots, t_m)$, and

$$\frac{\partial u}{\partial t_j} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_j}.$$



Implicit Differentiation

Suppose y is an implicit function of x given by $F(x, y) = 0$,
e.g., $F(x, y) = x^2 + y^2 - 1 = 0$ (unit circle), near a point (x_0, y_0) .

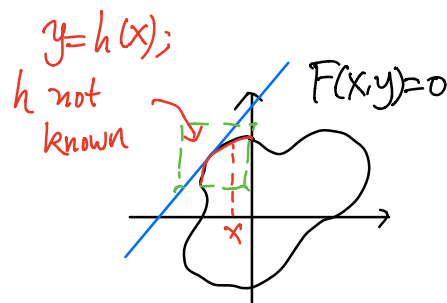
A level curve of F

Then near the point, $F(x, h(x)) = 0$

$\forall$ points on curve. Let $z = F(x, h(x))$.

Then $\frac{dz}{dx} = 0$. ① On the other hand, by

the chain rule,



$$\frac{dz}{dx} = F_x \cdot \frac{dx}{dx} + F_y \cdot \frac{dy}{dx} = F_x + F_y h'(x). \quad (2)$$

By ① & ②,

$$\frac{dy}{dx} = h'(x) = -\frac{F_x}{F_y} \quad (\text{if } F_y \neq 0).$$

14.9.8

THEOREM 8—A Formula for Implicit Differentiation Suppose that $F(x, y)$ is differentiable and that the equation $F(x, y) = 0$ defines y as a differentiable function of x . Then at any point where $F_y \neq 0$,

$$\frac{dy}{dx} = -\frac{F_x}{F_y}. \quad (1)$$

Example

Find the slope of the tangent line to the curve $x^3 + y^3 = 6xy$ at the point $(3, 3)$.

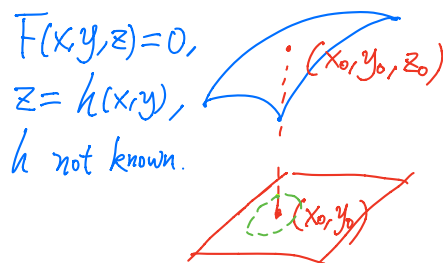
Sol: $F(x, y) = x^3 + y^3 - 6xy$; $F_x = 3x^2 - 6y$, $F_y = 3y^2 - 6x$

$$\frac{dy}{dx} \Big|_{(x,y)=(3,3)} = -\frac{F_x(3,3)}{F_y(3,3)} = -1.$$

Similarly, a surface $F(x, y, z) = 0$ may define z implicitly as

A level surface of F
a function of (x, y) near a point (x_0, y_0, z_0) , and $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ can be found similarly: near (x_0, y_0, z_0) ,

the surface is given by $F(x, y, h(x, y)) = 0$.



Let $w = F(x, y, h(x, y))$. Then $\frac{\partial w}{\partial x} = 0 = \frac{\partial w}{\partial y}$ near (x_0, y_0) .

Also, by the chain rule,

$$\frac{\partial w}{\partial x} = F_x \cdot \frac{\partial x}{\partial x} + F_y \frac{\partial y}{\partial x} + F_z \frac{\partial z}{\partial x} = F_x + F_z \frac{\partial z}{\partial x} \quad (=0),$$

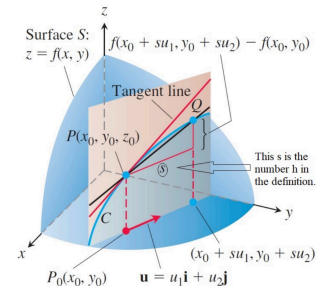
$$\frac{\partial w}{\partial y} = F_x \cdot \frac{\partial x}{\partial y} + F_y \frac{\partial y}{\partial y} + F_z \frac{\partial z}{\partial y} = F_y + F_z \frac{\partial z}{\partial y} \quad (=0),$$

So

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}, \quad \text{whenever } F_z \neq 0.$$

Exercise

Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $x^3 + y^3 + z^3 + 6xyz = 1$.



Directional Derivatives and Gradient Vectors

Q: Luigi is climbing a mountain and is now at horizontal position (x_0, y_0) , how fast will the altitude increase or decrease if he moves in the direction $\vec{u} = \langle u_1, u_2 \rangle$?

A: Directional derivative.

Definition

Let (x_0, y_0) be an interior point of the domain of a two-variable function f , and let $\vec{u}$ be a **unit vector** in $\mathbb{R}^2$. The **(directional) derivative of f at (x_0, y_0) in the direction of $\vec{u}$** , denoted by $D_{\vec{u}}f(x_0, y_0)$, is defined by

$$D_{\vec{u}}f(x_0, y_0) := \lim_{h \rightarrow 0} \frac{f(x_0 + hu_1, y_0 + hu_2) - f(x_0, y_0)}{h}.$$

In vector notation (which may generalize to n -variable functions) :

$$D_{\vec{u}} f(\vec{x}_0) = \lim_{h \rightarrow 0} \frac{f(\vec{x}_0 + h\vec{u}) - f(\vec{x}_0)}{h}.$$

Remark

Note that both partial derivatives are special cases of directional derivatives: $f_x = D_{\vec{u}_1} f$ and $f_y = D_{\vec{u}_2} f$, where $\vec{u}_1 = \langle 1, 0 \rangle$ and $\vec{u}_2 = \langle 0, 1 \rangle$.

Q: Given f , (x_0, y_0) and $\vec{u}$ (unit), how to compute $D_{\vec{u}} f(x_0, y_0)$?

Define $g(h) := f(\underbrace{x_0 + hu_1}_{x(h)}, \underbrace{y_0 + hu_2}_{y(h)})$. Then $g(0) = f(x_0, y_0)$, and

$$D_{\vec{u}} f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = g'(0).$$

On the other hand, by the chain rule,

$$\begin{aligned} g'(0) &= f_x(x(0), y(0)) \cdot x'(0) + f_y(x(0), y(0)) \cdot y'(0) \\ &= f_x(x_0, y_0) u_1 + f_y(x_0, y_0) u_2 = \nabla f(x_0, y_0) \cdot \vec{u}, \end{aligned}$$

where $\nabla f := \langle f_x, f_y \rangle$ is the **gradient vector**.

Theorem

If f is a differentiable two-variable function, then

$$D_{\vec{u}} f(x_0, y_0) = f_x(x_0, y_0) u_1 + f_y(x_0, y_0) u_2.$$

In other words,

$$D_{\vec{u}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \vec{u},$$

where $\nabla f := \langle f_x, f_y \rangle$.

Remark Def

The vector ∇f defined above is called the **gradient** or the **gradient vector** of f .

Example

Find the directional derivative of $f(x, y) := xe^y + \cos(xy)$ at the point $(2, 0)$ in the direction of $\vec{v} = \langle 3, -4 \rangle$.

Solution

-1.

In general, for an n -variable function $f(x_1, \dots, x_n)$,
 $\nabla f := \langle f_{x_1}, \dots, f_{x_n} \rangle$. If f is diff'able at a point $\vec{x}_0 \in \mathbb{R}^n$,
then it is still true that

$$D_{\vec{u}} f(\vec{x}_0) = \nabla f(\vec{x}_0) \cdot \vec{u}.$$

e.g. Consider $f(x, y) := \begin{cases} xy^2/(x^2+y^4), & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$

- Let $\vec{u}$ be any unit vector.

$$\begin{aligned} D_{\vec{u}} f(0, 0) &= \lim_{h \rightarrow 0} \frac{f(hu_1, hu_2) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h^3 u_1 u_2^2 / (h^2 u_1^2 + h^4 u_2^4)}{h} \\ &= \lim_{h \rightarrow 0} \frac{u_1 u_2^2}{u_1^2 + h^2 u_2^4} = \begin{cases} u_2^2 / u_1, & \text{if } u_1 \neq 0; \\ 0, & \text{if } u_1 = 0. \end{cases} \end{aligned}$$

- Hence, all directional derivatives exist for f at $(0, 0)$.

- But f is ^{even} not $\wedge$ cts at $(0, 0)$ as we have seen in Lecture 13,
let alone differentiable.

Geometric Meanings of Gradient Vectors

Q: When climbing mountain at the horizontal position (x_0, y_0) , which direction should you climb in order to go up the fastest? the slowest?

A: Look at gradient.

Geometric Meaning 1: direction of fastest increase and decrease.

Theorem

Let f be a differentiable function (with multiple variables), and let $\vec{x}_0$ represent an interior point of the domain of f . Then, among all directional derivatives $D_{\vec{u}}f(\vec{x}_0)$:

- ▶ The maximum value of $D_{\vec{u}}f(\vec{x}_0)$ is $|\nabla f(\vec{x}_0)|$, and it occurs when $\vec{u}$ is the direction of $\nabla f(\vec{x}_0)$.
- ▶ The minimum value of $D_{\vec{u}}f(\vec{x}_0)$ is $-|\nabla f(\vec{x}_0)|$, and it occurs when $\vec{u}$ is the direction of $-\nabla f(\vec{x}_0)$.

If $\nabla f(\vec{x}_0) \neq \vec{0}$ and $\vec{u}$ is orthogonal to $\nabla f(\vec{x}_0)$, then $D_{\vec{u}}f(\vec{x}_0) = 0$.

Remark

The theorem implies that at any point, a differentiable function increases the fastest in the direction of ∇f , and decreases the fastest in the direction of $-\nabla f$.

Exercise

Suppose that the temperature at a point (x, y, z) in space is given by $T(x, y, z) = 80/(1 + x^2 + 2y^2 + 3z^2)$, where temperature is measured in degrees Celsius and distance is measured in meters. In which direction does the temperature increase fastest at the point $(1, 1, -2)$? What is the maximum rate of increase?

Sol: · Want to find maximized $D_{\vec{a}} T(1,1,-2)$.

· $\nabla T(1,1,2) = \frac{5}{8} \langle -1, -2, 6 \rangle$.

· $\uparrow$ fastest in the direction $\vec{a} = \nabla T(1,1,2) / |\nabla T(1,1,2)|$.

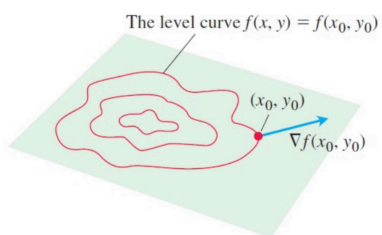
· $\text{Max } D_{\vec{a}} T(1,1,2) = |\nabla T(1,1,2)| = \frac{5}{8}\sqrt{41} \text{ } (^{\circ}\text{C/m})$

Geometric Meaning 2 : relation to level curves.

- Consider a level curve C of f given by $f(x,y) = k$, where f is diff'able.
- Suppose C has a parametrization $\vec{r}(t) = \langle x(t), y(t) \rangle$, $t \in I$.
- Then $f(x(t), y(t)) = k$, $\forall t \in I$.
- Differentiate both sides yields $\nabla f \cdot \vec{r}' = 0$, $\forall$ points on C .

Theorem

At any point (x_0, y_0) in the domain of a differentiable function f , if C is the level curve through (x_0, y_0) , then the gradient of f at (x_0, y_0) is perpendicular to the tangent vector to C at (x_0, y_0) .



This geometric fact allows us to describe the **tangent line to the level curve of f at a point (x_0, y_0)** using the equation

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0.$$

Example

Find an equation for the tangent line to the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$ at the point $(-2, 1)$.

Sol: • $f(x, y) = \frac{x^2}{8} + \frac{y^2}{2}$; $f'_x = \frac{x}{4}$, $f'_y = y$.
• $f'_x(-2, 1) = -\frac{1}{2}$, $f'_y(-2, 1) = 1$
• Tangent is $-\frac{1}{2}(x+2) + 1(y-1) = 0$.
 $(y = \frac{1}{2}x + 2 .)$

Following the approach for plane curves, we see that if f is defined on $D \subseteq \mathbb{R}^n$ and a curve C in D is parametrized by $\vec{r}(t) = \langle x_1(t), x_2(t), \dots, x_n(t) \rangle$, then

$$\frac{d}{dt} f(\vec{r}(t)) = \nabla f(\vec{r}(t)) \cdot \vec{r}'(t).$$

Properties of Gradients

Algebra Rules for Gradients

- | | | |
|----------------------------|--|---|
| 1. Sum Rule: | $\nabla(f + g) = \nabla f + \nabla g$ | |
| 2. Difference Rule: | $\nabla(f - g) = \nabla f - \nabla g$ | |
| 3. Constant Multiple Rule: | $\nabla(kf) = k\nabla f$ (any number k) | |
| 4. Product Rule: | $\nabla(fg) = f\nabla g + g\nabla f$ | Scalar multipliers on left of gradients |
| 5. Quotient Rule: | $\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}$ | |